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Particles of the Universe meets Distinti's Universe

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In this article, the most important equations in classical physics, including Einstein's energy-mass equivalence, Coulomb's law and Newton's laws, are compared to and validated against a set of force equations developed by independent researcher Robert Distinti. These results are further validated against Jeff Yee's equations from his wave model of a particle. It was found that all three sets of equations give equivalent results. Distinti's equations are slightly more detailed than the classical equations while Yee's equation offer much more detail. These extra details could point to a better and more complete understanding of the inner workings of nature. For this reason, further investigation into these alternate approaches is highly recommended.

Keywords: particles, waves, wave theory, aether, medium, electromagnetism, new electromagnetism

1. Introduction

Independent researcher, Robert Distinti, an electrical engineer for 30+ years, has developed a set of force equations that appear to encapsulate all known electromagnetic phenomenon. In this article, Distinti's equations are validated against the corresponding classical physics equations. This includes Einstein's energy-mass equivalence, Newton's second law of motion, Newton's universal law of gravitation, and the equation for centripetal force, which is also derived from Newton's laws of motion. The particle equivalent of each of these are mapped and calculated to the Coulomb force, Distinti's *New Induction*, the electron's magnetic moment and Distinti's *New Magnetism* respectively. These four equations are mapped in calculations using a single electron and then also derived to be a common equation using wave constants developed by independent researcher Jeff Yee. Finally, the relation between inductance and magnetism is explained as a conservation of energy and also derived to a common equation.

Through this process, it was realized that one approach or set of equations is not better than the other, in fact, they are equivalent. Although the standard equations are simple and more elegant, they do not provide sufficient detail to really understand the inner workings of nature. Distinti's equations provide a greater level of detail that is not apparent in the standard equations. A gateway model could be developed using Distinti's equations, but a better, more complete model may still exist. Yee's equations are not "simple and elegant", but offer the most detail leading to a more comprehensive and tangible model. Although these equation sets are equivalent, if one set is able to make a prediction that the other one doesn't than one can be better than the other. It is the contention of the authors that Distinti's equations are better than the standard equations. Yee's equations may be better yet. Further investigation is suggested.

2. Coulomb Force

The structure of particles, such as electrons, are described as standing waves of energy in Dr. Milo Wolff's paper *On the Spherical Wave Nature of Matter and the Origin of the Natural Laws*[1]. The Coulomb force is accurately modeled as a reflection of waves, diminishing in wave amplitude by the square of the distance from the source of the reflection, i.e., a particle. Longitudinal, traveling waves converge at the particle's core and are reflected to create standing waves. When amplitude diminishes such that the standing wave structure loses its form, longitudinal waves become traveling in form again. The Coulomb force can be modeled as the energy of a particle at distance, measured by the amplitude of the waves that are once again traveling waves as illustrated in 1.

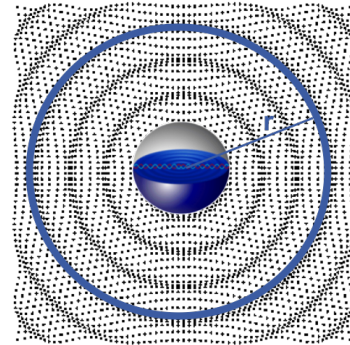


Figure 1. The electron's structure wave structure.

Wolff's work was expanded to include the energy of a particle's standing wave energy in *Particle Energy and Interaction*[2], including the derivation of Einstein's mass-energy equivalence ($E = mc^2$) in *Key Physics Equations and Experiments*[3]. New constants are introduced in the aforementioned papers for the calculations of particle energy. These constants, referred to hereafter as *wave constants*, are used in this paper as the common derivation be-

tween matter laws and electric/magnetic field laws.

The wave nature of particles has been witnessed in many experiments and has been explained for nearly as century as wave-particle duality[4]. By viewing a particle and its interactions as different wave forms, the convergence of different equations is possible, such as a particle's rest energy ($E = mc^2$) and the Coulomb force. Illustrations are used to view this behavior in both particle form (Figure 2) and wave form (Figure 3), but they are in reality, one in the same. Figure 2 describes the common visual that will be used throughout this paper for an electron that is spinning and traveling at some velocity less than the speed of light.

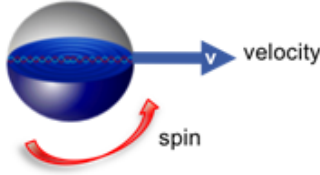


Figure 2. Particle structure.

Figure 3 illustrates the wave view of the same particle. In the center, concentric circles denote the standing wave structure of a particle. They are formed from longitudinal in-waves that are reflected to become out-waves. Beyond the radius of the standing wave structure (particle's radius), these longitudinal out-waves are traveling in form. As the particle reflects waves, it also causes spin. Some of the in-wave energy is used to create a transverse wave due to particle spin.

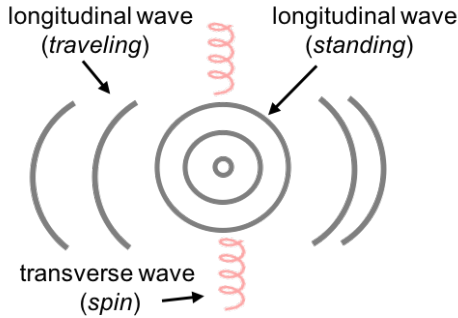


Figure 3. Wave structure.

These two views, particle view and wave view, are now shown for a particle at rest in Figure 4. The rest energy of a particle is calculated with Einstein's mass-energy equivalence ($E = mc^2$). The force of the particle on another particle (at distance) is calculated with the Coulomb force. Yet it will be shown that they are the same wave structure and converge to the same equation.

For the two equations to be equal, the same distance (r) needs to be considered for the energy measurement. According to the wave model of a particle, a single electron has energy beyond its particle radius (r_e), although wave amplitude diminishes from the source at the square of

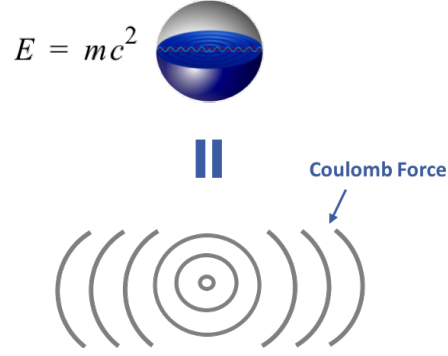


Figure 4. Coulomb force is electron's energy at a distance.

the distance when calculating the force. The left side of equation (1) combines the energy of a particle (mc^2) with the particle's radius (r_e) and a diminishing amplitude of the inverse square law ($1/r^2$). This is the force of a particle at rest. On the right side of (1) is the Coulomb force, where k_e is the Coulomb constant and q is elementary charge. Since force has both magnitude and direction, the unit vector $\hat{r}$ is included in these equations. Here, it is argued that these two equations are equivalent:

$$F = mc^2 \frac{r_e}{r^2} \hat{r} \equiv F_e = k_e q_1 q_2 \frac{1}{r^2} \hat{r} \quad (1)$$

To prove this equivalence, various distances are used in the calculation. The results are shown in Figure 5. The equation on the left of (1) is used for the *Particle View* for a single electron. The electron's mass (m) and radius (r_e) are used at four different distances. These results are then compared to the *Wave View* (Coulomb force) at the same distances for a single electron charge of q . As you can see from table in Figure 5, the forces for a particle at rest are identical. Distances are measured in meters (m) and forces in newtons (N). The maximum possible force, 29.1 N, occurs at the electron's radius.

The force of a particle at distance (r) and the Coulomb force are identical because the same wave equation can be derived from both the mass-energy equivalence equation (F) and the Coulomb force equation (F_e). Using *wave constants*, the two equations evaluate to the following equation:

$$F = \frac{4\pi p K_e^7 \lambda_l^2 c^2 \delta_e}{A_l \Delta_e} \frac{Q_1 Q_2}{r^2} \hat{r} \quad (2)$$

The derivation of both equations in (1) to a common wave equation (2) is found in Section 2 of The Relation of Electric and Magnetic Field Laws to Matter Laws[5].

3. New Induction

Robert Distinti proposes a revised equation for induction in *New Inductance Applications*[6]. In this section, we confirm his *New Inductance* equation by comparing it to Newton's second law of motion. Like the comparison of Coulomb's force to Einstein's mass-energy equivalence, it

Particle View

m = electron mass r_e = electron radius

Description	Distance	Particle Rest Force
Electron radius	2.82E-15	2.91E+01
Bohr radius	5.29E-11	8.24E-08
1 meter	1.00E+00	2.31E-28
Empire State Bldg	4.43E+02	1.18E-33

Wave View

q = single electron charge

Description	Distance	Coulomb Force
Electron radius	2.82E-15	2.91E+01
Bohr radius	5.29E-11	8.24E-08
1 meter	1.00E+00	2.31E-28
Empire State Bldg	4.43E+02	1.18E-33

Figure 5. Einstein's mass-energy equivalence is equal to Coulomb's energy (Coulomb force at distance).

is found that the two equations are equivalent when expressed in terms of wave constants.

Figure 6 depicts the particle view and the wave view of a particle in motion. The force of an object in motion is calculated using Newton's second law ($F = ma$). In the wave view of a particle, the wavelength of the particle decreases on the leading edge in the direction of motion and increases on the trailing edge. This is due to the Doppler effect[10]. Amplitude changes as a result of constructive wave interference from other particles in proximity.

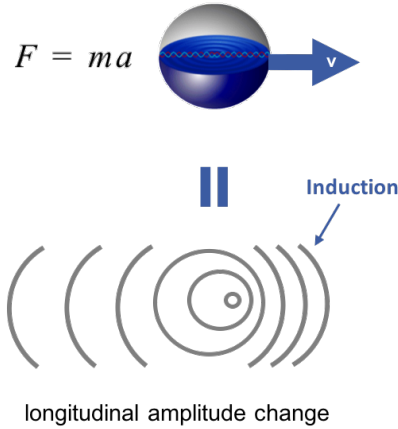


Figure 6. Induction is a change in amplitude (motion)

The two equations, corresponding to Figure 6 are as follows:

$$F = ma \equiv F_i = K_M \frac{q_1 q_2}{|r|} a \quad (3)$$

These two equations are found to be identical when derived in terms of wave constants (equation 4). The calculations are also identical as shown in Figure 7. To test the equivalence, a single electron is used, hence, the mass of the electron is used for the mass parameter, m , the classical electron radius is used for the distance parameter, r , and elementary charge is used for the charge parameters, q_1 and q_2 . The parameter, K_m is equal to the magnetic constant μ_0 divided by 2π .

The table in Figure 7 compares Newton's second law (particle view) and Distinti's *New Induction* (Wave View) for four different acceleration values. As you can see, the

force calculations are identical. The maximum force of 29.1 newtons for a single electron occurs at the speed of light squared divided by the electron's radius (c^2/r_e). Since force equations vary with distance, it will be shown throughout this paper that the transition point from standing waves to traveling waves (electron radius) is the point at which forces are equal (i.e., opposing forces balance).

Since the equation for induction varies by distance (r), where Newton's second law does not, the two equations are converted to energy equations. Using *wave constants*, both equations from (3) evaluate to the same energy equation as follows:

$$E = \frac{4\pi p K_e^7 \lambda_l^2 c^2 \delta_e}{A_l \Delta_e} \frac{Q_1 Q_1}{r^2} a \quad (4)$$

The derivation of both equations to a common wave equation is found in Section 3 of *The Relation of Electric and Magnetic Field Laws to Matter Laws*[5].

4. New Magnetism

Robert Distinti also proposed a revised equation for magnetism in *New Magnetism*[7]. In this article, the equation for *New Magnetism* shown to be equivalent to the equation used for the calculation of an object's centripetal force. Figure 8 depicts a spinning particle and how it generates a new transverse wave.

A particle naturally spins, even at rest. The calculation for the magnetic moment, when at rest, will be shown in the last section. Electromagnetism, generated by the current flow of electrons at velocity (v) was modeled by Distinti in the *New Magnetism* equation. A modified version of the point charge form of this equation is shown on the right side of equation (5). On the left side of (5) is the standard centripetal force equation used for objects with mass. Once again, it will be shown that these two equations are equivalent as they can be derived from identical wave constant equations. The calculations are also identical as can be seen in Figure 9. The two equations, corresponding to Figure 8 are as follows:

$$F_c = \frac{mv^2}{r} \hat{r} \equiv F_m = K_M \frac{q_1 q_2}{r^2} v^2 \hat{r} \quad (5)$$

On the left side, the centripetal force equation (F_c) is used to calculate a single electron's force at various dis-

Particle View

m = electron mass

Description	Acceleration	Particle Motion Force
Electron acceleration	3.50E+15	3.19E-15
1 meter / second ^2	1.00E+00	9.11E-31
Car acceleration	4.00E+00	3.64E-30
c^2 / electron radius	3.19E+31	2.91E+01

Wave View

r = electron radius

Description	Acceleration	Induction Force
Electron acceleration	3.50E+15	3.19E-15
1 meter / second ^2	1.00E+00	9.11E-31
Car acceleration	4.00E+00	3.64E-30
c^2 / electron radius	3.19E+31	2.91E+01

Figure 7. Newton's second law is equal to New Induction for a single electron at the electron's radius.

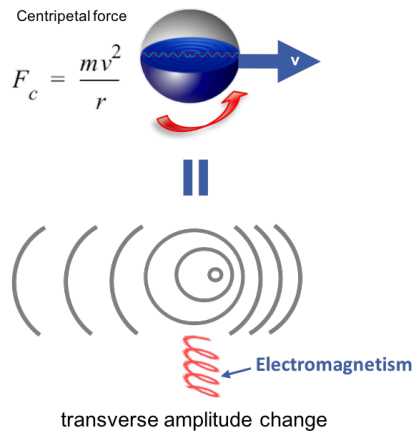


Figure 8. Electromagnetism is a transverse wave change.

tances and velocities. On the right side, elementary charge is used to calculate the magnetic force (F_m). As you can see from the table in Figure 9, centripetal force is equal to magnetic force for all test cases. In particular, when classical electron radius, r_e , and speed of light, c , are used for radius and velocity respectively, the force evaluates to 29.1 Newtons for both equations.

As in the previous cases, these two equations can be derived from the same wave constant equation:

$$E = \frac{4\pi p K_e^5 \lambda_l \delta_e}{A_l \Delta_e} Q_1 Q_1 v^2 \quad (6)$$

The derivation of both equations to a common wave equation is found in Section 4 of *The Relation of Electric and Magnetic Field Laws to Matter Laws*[5].

5. Faraday's Law

The derivation of *New Inductance* and *New Magnetism* to wave constant form provides a better understanding of the relationship between inductance and magnetism, which was first captured by Michael Faraday in Faraday's Law, and which also became one of the four important equations from Maxwell's equations. Inductance and magnetism can also be related through the conservation of energy principle.

When a current is induced, electrons will flow due to a change in wave amplitude, i.e., particles will move to the point of minimal wave amplitude. In Figure 10, the top (Induction) illustrates a wave view of a particle in motion. Due to the motion of the particle, it causes greater constructive wave interference and thus greater amplitude. A particle in the direction of motion will be forced to move as a result of this amplitude difference. The process may repeat itself as electrons continually do this down a wire, causing a current.

On the bottom of 10 is the wave view of magnetism. The particle's motion has an effect on the spin. A faster velocity causes a faster spin, increasing the amplitude of the transverse (spin) wave.

In most materials, the alignment of particle spin is often canceled as particles move to minimize wave amplitude. Ferromagnetic materials are an exception as they may have a greater number of same-spin particles and can be aligned to produce a magnetic force, even at rest. There are two directions for spin, which may be thought of as clockwise (CW) and counter-clockwise (CCW). Two particles of the same spin (CW + CW or CCW + CCW) will be constructive and repel each other. Two particles of opposite spin (CW + CCW) will be destructive and attract each other.

In Figure 11, two particles are shown. Let's assume that the top particle is an electron and the bottom particle is a proton. When the particles are at rest, the spin waves cancel (CW + CCW), and there is no force. When the electrons are in motion, each electron jumps to the next proton in the sequence, thus, the electron has velocity and the proton does not. While each particle spins, the rotation causes various transverse waves to form, noted by the helix shaped field lines. Figure 11 shows how the amplitude is affected by the motion. The greatest amplitude difference will be directly between the particles as it is perpendicular to the motion of the electron. Particles will move to the point of minimal amplitude which causes a force in the direction perpendicular to the current.

This model fits well with the Stern-Gerlach experiment which demonstrates the two possible spins for the electron. When traveling at velocity in a magnetic field, the force will be perpendicular to the direction of motion. In Figure 12, two electrons travel through a magnetic field. The

Particle View

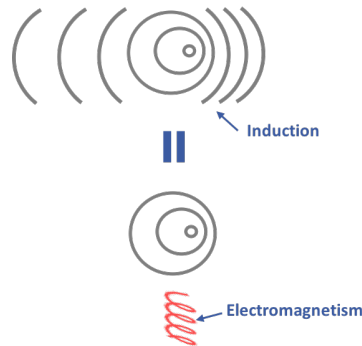
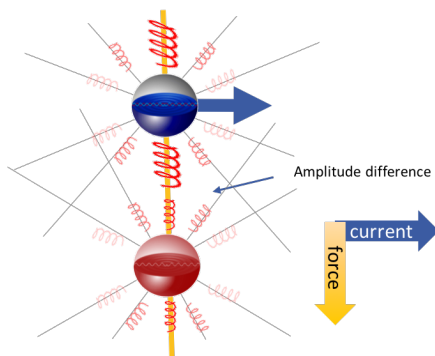
m = electron mass r = electron radius

Description	Velocity	Centripetal Force
Electron drift speed	2.50E-04	2.02E-23
1 m/s	1.00E+00	3.23E-16
car highway speed	2.90E+01	2.72E-13
speed of light	299,792,458	2.91E+01

Wave View

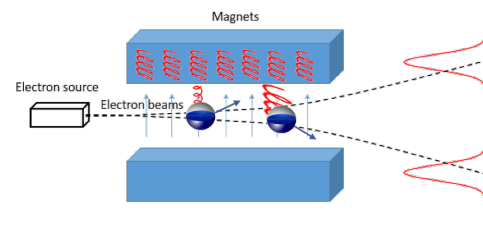
q = single electron charge r = electron radius

Description	Velocity	Magnetic Force
Electron drift speed	2.50E-04	2.02E-23
1 m/s	1.00E+00	3.23E-16
car highway speed	2.90E+01	2.72E-13
speed of light	299,792,458	2.91E+01

Figure 9. Centripetal force is equal to New Magnetism for a single electron at the electron's radius.**Figure 10.** Induction and electromagnetism**Figure 11.** Electron motion and force direction

first emitted electron (on the right) has spin in the same direction as top magnet and is constructive. It would be destructive with the bottom magnet (not shown for purpose of simple illustration). The result is a greater amplitude between the top magnet and the electron, forcing the electron down. The second electron emitted from the source (on the left) has spin in opposite direction of the top magnet and is destructive. There is now a greater amplitude in the direction of the bottom magnet forcing the electron up.

The wave energy forms of the equations for induction and magnetism are derived from (4) and (6) respectively to show that these are indeed equal due to the conservation of energy as expressed in the equivalent equations, (7 and 8). The equations (4) and (6) are used for the calculations

**Figure 12.** Stern-Gerlach experiment.

in Figure 14.

$$E_i = \frac{4\pi p K_e^7 \lambda_l^2 \delta_e}{A_l \Delta_e} Q_1 Q_2 a \quad (7)$$

$$E_m = \frac{4\pi p K_e^5 \lambda_l \delta_e}{A_l \Delta_e} Q_1 Q_2 v^2 \quad (8)$$

In the table in Figure 14, the wave equation energy form of the induction and magnetism equations are used for various calculations of velocities and distances (r) of a single electron. These are shown in columns 4 and 5 respectively. As you can see, the energy calculations are equal. In columns 6 and 7, the forces are calculated. Not surprisingly, the forces are equal at the electron's radius (in bold) but differ at other distances. This is due to inductance decreasing at distance (r), and new magnetism at the square of distance. At the limit, which is the speed of light and the electron radius, the energy value calculated is equal to the electron's rest energy (8.19×10^{-14} joules), and the maximum force is once again 29.1 newtons.

Section 5 of *The Relation of Electric and Magnetic Field Laws to Matter Laws* describes how energy is conserved and how inductance energy (E_i) can be derived to be magnetic energy (E_m) in a common wave equation[5].

6. Magnetism and Newton's Law of Gravity

Distinti's *New Magnetism* equations support the calculations of electrons in motion. Even at rest, the electron spins and contains a magnetic moment. This can be related to yet another one of Newton's equations, i.e., the universal law

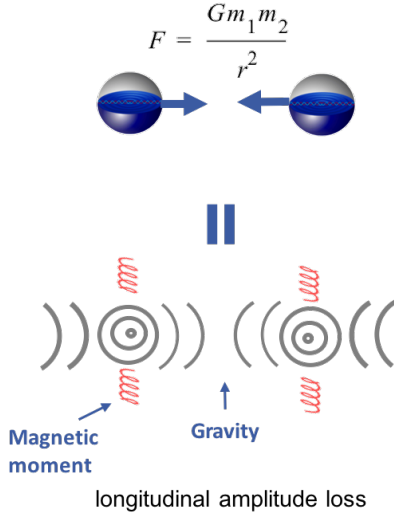


Figure 13. Gravity is equal to magnetic moment energy

of gravitation. In Figure 13, two electrons are illustrated with a gravitational effect.

This effect is very slight, as it does not cause an attraction at the particle-level due to the Coulomb force. Gravity occurs in large bodies because it requires the cancellation of longitudinal wave forces in atoms (protons and electrons) before the effect can be seen as attractive. Nevertheless, the force can be calculated using Newton's law as shown on the top of the figure (particle view). On the bottom (wave view), the spin wave for a particle at rest is described as the magnetic moment. This is the rest spin energy of a single electron. Also illustrated is the loss of wave amplitude as *in-waves* convert to *out-waves*. This is visually described by thicker lines (greater amplitude) for *in-waves* and thinner lines (lesser amplitude) for *out-waves*. The cause is related to the conservation of energy. Some of the *in-wave* energy is used for particle spin, thus reducing *out-wave* energy. This causes a shading effect between two particles where amplitude is lower, causing particles to move to the point of minimal amplitude, which is the cause of gravity. Refer to the *Forces* paper [8] for more details and calculations on this behavior.

The loss of energy between in-waves and out-waves is identical to the energy of the electron's magnetic moment. In other words, the energy for gravity is the spin energy of an electron at rest. The gravitational energy can be calculated using two different methods, as shown in equations (9) and (10) where (9) is the gravitational energy of an electron particle at the electron's classical radius using Newton's gravitational law respectively and (10) is electron energy multiplied by the gravitational coupling constant for the electron (α_{G_e} = gravitational strength divided by electric strength).

Equation (11) is an expression of two magnetic moment energies of an entire electron (measured at the electron's radius). Two energies are required due to two transverse

waves being created and traveling in opposite directions. The details of each constant are explained in *The Relation of Electric and Magnetic Field Laws to Matter Laws*. Notice that the energies of all three equations evaluate to a very small 1.97×10^{-56} joules for a single electron.

$$E = \frac{Gm_e^2}{r_e} = 1.97 \times 10^{-56} [J] \quad (9)$$

$$E = m_e c^2 \alpha_{G_e} = 1.97 \times 10^{-56} [J] \quad (10)$$

$$E_m = 2 \frac{\mu_0}{\delta_e} \left(\frac{K_e q_p \mu_B}{r_e} \right)^2 O_e = 1.97 \times 10^{-56} [J] \quad (11)$$

Once again, all three equations can be derived from identical equations in wave constant form which explains why the calculations in (9), (10) and (11) are equal. The two forms for the gravitational energy of a single electron (E_g) and the magnetic energy of a single electron at rest (E_m) is shown in (12).

$$E_g = E_m = \frac{2p\lambda_l^4 c^2 \delta_e}{3K_e^{15} A_l^4} Q_1 Q_2 \quad (12)$$

The derivation of all equations to a common wave equation is found in Section 6 of The Relation of Electric and Magnetic Field Laws to Matter Laws[5].

7. Summary of Forces

In all of the calculations for forces and energy in this paper, there was a limit for energy that was equal to the electron's energy (E_e) and a force of 29.1 newtons that occurs at the electron's radius (r_e). Again, this occurs because all of the calculations were performed for single electron particles. This leads to a simplified description of forces. Forces are based on the energy of a group of particles (Q_1) that have diminishing traveling wave amplitude (energy) based on a distance (r). The electron particles have a transition point where standing waves become traveling waves at the electron's radius (r_e). The energy at this point is described by (13). **The common terms for electron energy and radius** are used instead of wave constant terms for simplicity. They can be expanded in form to derive the wave constant equation formats.

$$E = E_e \frac{Q_1 r_e}{r} \quad (13)$$

A force on a group of particles (Q_2) is related to the energy from the equation above, at a distance (r) by the following:

$$F = E \frac{Q_2}{r} \hat{r} \quad (14)$$

Since this is now a force equation, the unit vector $\hat{r}$ is shown. It should be noted that in equations (13) and (14),

Description	Velocity (final)	Acceleration	Energy (induction)	Energy (magnetic)	Force Distance (radius)	Force (induction)	Force (magnetic)
Electron drift speed - electron radius	2.50E-04	2.22E+07	5.69E-38	5.69E-38	2.82E-15	2.02E-23	2.02E-23
Electron drift speed - random r	2.50E-04	2.22E+07	5.69E-38	5.69E-38	2.00E+00	2.85E-38	4.01E-53
Car highway speed - electron radius	2.90E+01	2.98E+17	7.66E-28	7.66E-28	2.82E-15	2.72E-13	2.72E-13
Car highway speed - random r	2.90E+01	2.98E+17	7.66E-28	7.66E-28	3.00E+00	2.55E-28	2.40E-43
Electron acceleration - electron radius	2.50E-04	2.22E+07	5.69E-38	5.69E-38	2.82E-15	2.02E-23	2.02E-23
Electron acceleration - random r	2.50E-04	2.22E+07	5.69E-38	5.69E-38	4.00E+00	1.42E-38	1.00E-53
Car acceleration - electron radius	1.06E-07	4.00E+00	1.03E-44	1.03E-44	2.82E-15	3.64E-30	3.64E-30
Car acceleration - random r	1.06E-07	4.00E+00	1.03E-44	1.03E-44	5.00E+00	2.05E-45	1.16E-60
Limits	299,792,458	3.19E+31	8.19E-14	8.19E-14	2.82E-15	2.91E+01	2.91E+01

Figure 14. Induction and magnetism. Energy is always equal and conserved. Forces are equal at electron's radius.

Q_1 and Q_2 are a dimensionless count of particles in these equations, not charge. All of the forces described in this paper (magnetic, electric and gravitational forces) share this common structure. In fact, two equations for the strong force (gluon interaction and atomic orbital interaction) also share the same structure but are not covered in this paper. Instead, these equations can be found in the *Forces* paper and *Atomic Orbitals* paper [9].

The Coulomb force is the fundamental force based on longitudinal waves. It is the substitution of (13) into (14) and expressed in simple terms in (15).

$$F = E_e \frac{Q_1 r_e}{r} \frac{Q_2}{r} \hat{r} \quad (15)$$

The remaining forces correspond to a change in wave form or amplitude, yet the principle always remains the same. Particles move to the point of minimal amplitude. The following are the differences from the Coulomb (electric force), also found in visual format with simplified force equations in Figure 13:

1. • **Electromagnetism:** change in energy from rest where c^2 is replaced by v^2 in E_e .
2. • **Gravitation:** change in longitudinal energy where α_{Ge} is applied to wave amplitude. The energy loss is due to particle spin and is converted to the transverse (spin) energy.
3. • **Strong Force:** change in energy from kinetic energy, now stored, where α_e is applied to wave amplitude. *Note wave amplitude increases by the inverse of wave structure constant which is why it appears in the denominator.*
4. • **Orbital Force:** change in energy as a wave passes linearly through two particles affected by the strong force where α_e is applied to wave amplitude. It is now squared because it passes through two particles before affecting the orbiting electron, now decreasing at the cube of distance. See *Atomic Orbitals* (reference) for explanation.

8. Discussion

"There is always another way to say the same thing, that doesn't look at all like the way you said it before". Richard Feynman [11]. The purpose of this exercise was to compare Robert Distinti's *New Magnetism* and *New Induction* standard equations from main stream physics. Using equations derived from wave constants from Jeff Yee's "Particles of the Universe", it was found that Distinti's equations are in fact equivalent to the mainstream equations. For example, it was found that $F = ma$ is equivalent to $F = (K_M \frac{q_1 q_2}{r})a$. Since these two equations evaluate to the same force, we can see that one equation is not necessarily better than the other, however, one does offer more detail than the other. For instance, $F = ma$ is useful for calculating the force of a known mass in motion, but it does not give us any ideas as to what is the cause of mass. Distinti's equation gives us some clues as to the root cause of mass. From equation (3), it is clear to see that $m = (K_M \frac{q_1 q_2}{r})$. This seems to be telling us that the cause of mass has something to do with charges at distance. Yee's equation in energy form, (4) offers quite a bit more detail leading to a more comprehensive explanation for mass. If the purpose of a model is to understand the universe, then the more detail you can give, the closer you will likely be to the truth. Many models can be used to explain the same thing. One model is not necessarily better than the other. That said, if one model can make a prediction that the other one can't, then technically, that model is better. We may never know the exact correct model for the universe, but the more we study it, the closer we will get.

9. Conclusion

Robert Distinti's equations for electromagnetism include the commonly known 1) Coulomb force, and new equations for 2) New Inductance and 3) New Magnetism. All three of these equations were mapped to known equations for classical physics, illustrating that particles are waves and continue to be beyond the particle's borders.

The Coulomb force was mapped to the rest energy of a particle, using Einstein's mass-energy equivalence and extending the particle energy beyond the border of the electron. The *New Inductance* force was mapped to New-

Force	Visual	Description	Equation
Coulomb's Law (Electric Force)		Energy a distance (r) from electrons (Q ₁). Waves have passed through standing waves of electron (classical radius) and are traveling again. Interfere (+ or -) with distant electrons (Q ₂).	$F = E_e \frac{Q_1 r_e}{r} \frac{Q_2}{r}$
Distinti's Magnetism (Magnetic Force)		Energy converted from longitudinal waves to transverse (spin) from electrons (Q ₁). Constructive or destructive spin waves at a distance r from distance electrons (Q ₂).	$F = m_e v^2 \frac{Q_1 r_e}{r} \frac{Q_2}{r}$
Newton's Gravitation (Gravitational Force)		Energy difference a distance (r) due to lost energy (α _{ge}) from particle spin (Q ₁). The difference produces a shading effect on Q ₂ and is attractive.	$F = E_e \frac{Q_1 r_e \alpha_{ge}}{r} \frac{Q_2}{r}$
Strong Interaction (Strong Force)		Energy stored from particle kinetic energy (α _e) of Q ₁ and Q ₂ . When at standing wave nodes (stable) at a distance (r), particles are attractive. Energy becomes a stored "gluon".	$F = E_e \frac{Q_1 r_e}{\alpha_e r} \frac{Q_2}{r}$
Strong Interaction (Orbital Force)		Energy a distance (r) from the proton after wave passes through two particles (Q ₁) with gluon energy and goes through energy change. The force decreases at the cube of distance.	$F = E_e \left(\frac{Q_1 r_e}{\alpha_e r} \right)^2 \frac{Q_2}{r}$

Q (number) of Particles
 Wave (Energy)
 Particle Distance
 Electron Radius
 Proton

Figure 15. A summary of forces and the relation to a single electron's energy and the electron's classical radius.

ton's second law of motion. The *New Magnetism* force was mapped to the centripetal force equation for objects as well as from Newton's laws of motion. All three equations match for a single electron when the radius measured is the electron's classical radius. These equations were then derived to wave constant form and shown to be equal using the conservation of energy principle.

The conservation of energy was also applied to the relationship between inductance and magnetism, otherwise known as Faraday's Law. Again, this was shown to be equivalent in calculations at the electron's radius and derived in wave constant form to be equivalent.

Finally, a scenario not covered under Distinti's equations was mapped to yet another equation from Newton. An electron has spin, even at rest. This energy, calculated using the magnetic moment of a single electron, was mapped to Newton's law of universal gravitation. Once again, for a single electron at the electron's radius, the force and energy calculations are equal.

Energy is always conserved and thus can be modeled mathematically, tying together the aforementioned classical equations typically used for objects (and now particles), and mapping these to equations typically used for fields of electrical and magnetic energy. It is found that they are one in the same. As an electron moves, it affects other particles with both its reflective, longitudinal waves and its spinning, transverse waves. This is the basis for the electrical and magnetic forces respectively. This shows the true nature of particles, in particular the electron, as a formation of waves. It also provides a validation of Distinti's new equations for inductance and magnetism.

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the relationship between classical laws of matter and the field laws for electricity and magnetism would not have been possible.

REFERENCES

- Wolff, M. "On the Spherical Wave Nature of Matter and the Origin of the Natural Laws." Online: <http://www.spaceandmotion.com/Wolff-Wave-Structure-Matter.htm> (Accessed October, 2017)
- Yee, J. "Particle Energy and Interaction: Explained and Derived by Energy Wave Equations." Vixra.org, 1408.0224 (2018).
- Yee, J. "Key Physics Equations and Experiments: Explained and Derived by Energy Wave Equations." Vixra.org, 1705.0101 (2018).
- Bahr, Benjamin, Boris Lemmer, and Rina Piccolo. "Wave-Particle Duality." In *Quirky Quarks*, pp. 142-145. Springer, Berlin, Heidelberg, 2016.
- Yee, J., Gardi, L. "The Relation of Electric and Magnetic Field Laws to Matter Laws." Vixra.org, 1806.0071 (2018).
- Distinti, R. "New Inductance Applications." Online: <http://www.distinti.com/distinti/ne/ne.htm> (Accessed March, 2018).
- Distinti, R. "New Magnetism." Online: <http://www.distinti.com/distinti/ne/ne.htm> (Accessed March, 2018).
- Yee, J. "Forces: Explained and Derived by Energy Wave Equations." Vixra.org, 1606.0112 (2018).
- Yee, J., Zhu, Y. and Zhou, G. "Atomic Orbitals: Explained and Derived by Energy Wave Equations." Vixra.org, 1708.0146 (2018).
- Possel, M. "Waves, motion and frequency: the Doppler effect." *Einstein Online*, Vol 5. Max Planck Institute for Gravitational Physics, Potsdam, Germany, (Accessed March, 2018).
- Feynman, Richard P. "The development of the space-time view of quantum electrodynamics." *Science* 153, no. 3737 (1966): 699-708.